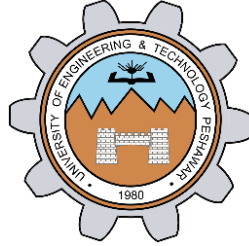


Database Development using MySQL

LAB # 04



Spring 2024

CSE-403L Database Management Systems Lab

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“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Date:

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Objectives:

This lab aims at the understanding of:

- An Example SQL Database Management System i.e. MySQL
- Using MySQL Command Line in Windows
- MySQL Command Categories
- Use of MySQL Commands for a Real-World Problem

Task 4.1:

Q1: What is DDL, DML, TCL, and DCL? Explain in your own words. Also, list few commands in each language.

Ans: DDL (Data Definition Language):

DDL deals with defining and modifying the structure of database objects (like tables, indexes, views, etc.). It doesn't manipulate data itself.

Common DDL commands: CREATE, DROP, ALTER, TRUNCATE, COMMENT, RENAME etc

DML (Data Manipulation Language):

DML is all about manipulating data within schema objects.

Key DML command: SELECT

DCL (Data Control Language):

DCL focuses on controlling access to data.

Common DCL commands: GRANT, REVOKE

TCL (Transaction Control Language):

TCL manages transactions (groups of SQL statements that are executed together).

Key TCL commands: COMMIT, ROLLBACK, SAVEPOINT

Q3: Evaluate yourself and write a note about your understanding of this lab.

Ans: I learned all the terminologies related to SQL Database Management System. I learned how to create a database when relational schema is given. I also learned how to populate and manipulate a database using DBMS.

Task 4.2:

What is the difference between SQL and MySQL? Why is MySQL used? What are its features?

Ans:

SQL:

SQL is short for Structured Query Language. SQL is the standard language to maintain and manage a database. Thus, besides a few minor syntax changes, most of the databases' queries for retrieving, adding, or manipulating data are based on the standard SQL syntax.

MySQL:

MySQL is a relational database management system developed and released back in 1995. It was developed and sponsored by MySQL AB, which was then acquired by Sun Microsystems, which you know today as Oracle Corporation. MySQL is a combination of 'My' (co-founder's daughter) and 'SQL.'

Why use MySQL: MySQL's versatility, maturity, open-source nature, and extensibility contribute to its widespread adoption in the database world. Whether you're building a small web application or managing a large-scale system, MySQL remains a reliable choice!

MySQL Features:

1) Easy to Use 2) Scalability 3) Performance 4) Reliability 5) Compatibility 6) Open Source

Task 4.3:

What is database engine? What purpose does it serve? How many types of engines are supported by MySQL? Which database engine is most commonly used and why?

Ans:

Database Engine:

A database engine is the underlying system that a database uses to function. Many different technologies rely on internal "engines," which are the fundamental building blocks on which they operate.

Purpose of Database Engines:

- Data Storage: Database engines store data efficiently on disk or in memory.
- Data Retrieval: They retrieve data based on queries (SELECT statements).
- Transaction Management: They handle transactions (groups of SQL statements executed together).
- Concurrency Control: Ensures data consistency when multiple users access the database simultaneously.
- Indexing: Optimizes data retrieval by creating indexes on columns.

Types of Database Engines Supported by MySQL:

- MySQL supports several storage engines, each with specific features and use cases:
 - InnoDB: The default and most general-purpose engine. Supports transactions, row-level locking, and foreign keys. Recommended for most tables.
 - MyISAM: A non-transactional engine known for its speed. Suitable for read-heavy workloads but lacks features like transactions and foreign keys.
 - MEMORY (HEAP): Stores data in memory for fast access. Useful for temporary tables or caching.
 - ARCHIVE: Optimized for storing large amounts of historical data with minimal space usage.
 - BLACKHOLE: Acts like /dev/null—anything written to it disappears.
 - MERGE: Combines multiple identical MyISAM tables into a single virtual table.
 - FEDERATED: Allows data from remote servers to be accessed as if it were local.
 - EXAMPLE: A dummy engine for testing and documentation purposes.

Most Commonly Used Engine:

- InnoDB is the go-to choice for most applications due to its reliability, ACID compliance, and support for transactions.
- It provides row-level locking, crash recovery, and foreign key constraints.
- InnoDB is well-suited for web applications, e-commerce platforms, and any scenario where data integrity and reliability are critical.

Task 4.4:

Specify at least fifteen (15) or more different data types supported by MySQL. Provide the description with at least one example.

Ans:

Integer (INT): Represents whole numbers (positive or negative).

Example: `age INT` for storing a person's age.

Decimal (DECIMAL): Stores fixed-point numbers with specified precision and scale.

Example: `price DECIMAL(10, 2)` for storing a product price with two decimal places.

Date (DATE): Stores dates in the format 'YYYY-MM-DD'.

Example: `birth_date DATE` for storing a person's birthdate.

Time (TIME): Represents time values in the format 'HH:MM:SS'.

Example: `start_time TIME` for recording event start times.

Datetime (DATETIME): Combines date and time in the format 'YYYY-MM-DD HH:MM:SS'.

Example: `created_at DATETIME` for timestamping record creation.

Timestamp (TIMESTAMP): Similar to DATETIME but automatically updates on data modifications.

Example: `last_modified TIMESTAMP` for tracking changes.

Char (CHAR): Fixed-length string with a specified maximum length.

Example: `country_code CHAR(2)` for storing country codes.

Varchar (VARCHAR): Variable-length string with a maximum length.

Example: `product_name VARCHAR(255)` for product names.

Text (TEXT): Stores large text data (longer than VARCHAR).

Example: `description TEXT` for product descriptions.

Boolean (BOOL or BOOLEAN): Represents true or false values.

Example: `is_active BOOLEAN` for tracking active status.

Enum (ENUM): Defines a set of allowed values.

Enum (ENUM): Defines a set of allowed values.

Example: `gender ENUM('Male', 'Female', 'Other')` for gender selection.

Set (SET): Similar to ENUM but allows multiple values.

Example: `interests SET('Reading', 'Traveling', 'Cooking')` for hobbies.

Binary (BINARY): Stores binary data (e.g., images, files).

Example: `profile_picture BINARY` for user avatars.

Blob (BLOB): Large binary object for storing unstructured data.

Example: `resume BLOB` for job applicants' resumes.

JSON (JSON): Stores JSON-formatted data.

Example: `user_data JSON` for flexible user attributes.

Task 4.5:

```
mysql> show tables;
+-----+
| Tables_in_lab4 |
+-----+
| entry          |
| member         |
| team           |
| tournament     |
| type           |
+-----+
5 rows in set (0.01 sec)
```

```
mysql> describe type;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| type  | varchar(10)   | NO   | PRI | NULL    |       |
| fee   | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> show create table type;
+-----+-----+
| Table | Create Table
+-----+-----+
| type  | CREATE TABLE `type` (
  `type` varchar(10) NOT NULL,
  `fee` int DEFAULT NULL,
  PRIMARY KEY (`type`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
+-----+-----+
1 row in set (0.01 sec)

mysql> _
```

```
mysql> describe tournament;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| TourID     | int           | NO   | PRI | NULL    | auto_increment |
| Tournamen | varchar(50)   | YES  |     | NULL    |                |
| TourType   | varchar(10)   | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
```

```
mysql> show create table tournament;
+-----+-----+
| Table      | Create Table
+-----+-----+
| tournament | CREATE TABLE `tournament` (
  `TourID` int NOT NULL AUTO_INCREMENT,
  `Tournamen` varchar(50) DEFAULT NULL,
  `TourType` varchar(10) DEFAULT NULL,
  PRIMARY KEY (`TourID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> describe team;
```

Field	Type	Null	Key	Default	Extra
Teamname	varchar(30)	NO	PRI	NULL	
practicenight	varchar(50)	YES		NULL	
manager	int	YES	MUL	NULL	

```
3 rows in set (0.01 sec)
```

```
mysql> show create table team;
```

Table	Create Table
team	<pre>CREATE TABLE `team` (`Teamname` varchar(30) NOT NULL, `practicenight` varchar(50) DEFAULT NULL, `manager` int DEFAULT NULL, PRIMARY KEY (`Teamname`), KEY `manager` (`manager`), CONSTRAINT `team_ibfk_1` FOREIGN KEY (`manager`) REFERENCES `member` (`memberID`)) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci</pre>

```
1 row in set (0.00 sec)
```

```
mysql>
```

```
mysql> describe member;
```

Field	Type	Null	Key	Default	Extra
memberID	int	NO	PRI	NULL	
FirstName	varchar(20)	YES		NULL	
LastName	varchar(20)	YES		NULL	
memberType	varchar(20)	YES	MUL	NULL	
Phone	int	YES	UNI	NULL	
Handicap	int	YES		NULL	
JoinDate	date	YES		NULL	
coach	int	YES	MUL	NULL	
gender	varchar(10)	YES		NULL	
team	varchar(20)	YES	MUL	NULL	

```
10 rows in set (0.00 sec)
```

```
| member | CREATE TABLE `member` (
  `memberID` int NOT NULL,
  `FirstName` varchar(20) DEFAULT NULL,
  `LastName` varchar(20) DEFAULT NULL,
  `memberType` varchar(20) DEFAULT NULL,
  `Phone` int DEFAULT NULL,
  `Handicap` int DEFAULT NULL,
  `JoinDate` date DEFAULT NULL,
  `coach` int DEFAULT NULL,
  `gender` varchar(10) DEFAULT NULL,
  `team` varchar(20) DEFAULT NULL,
  PRIMARY KEY (`memberID`),
  UNIQUE KEY `Phone` (`Phone`),
  KEY `memberType` (`memberType`),
  KEY `team` (`team`),
  KEY `coach` (`coach`),
  CONSTRAINT `member_ibfk_1` FOREIGN KEY (`memberType`) REFERENCES `type` (`type`),
  CONSTRAINT `member_ibfk_2` FOREIGN KEY (`team`) REFERENCES `team` (`Teamname`) ON UPDATE SET NULL,
  CONSTRAINT `member_ibfk_3` FOREIGN KEY (`coach`) REFERENCES `member` (`memberID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
```

```
mysql> describe entry;
```

Field	Type	Null	Key	Default	Extra
memberID	int	YES	MUL	NULL	
TourID	int	YES	MUL	NULL	
year	year	YES		NULL	

3 rows in set (0.00 sec)

```
mysql> show create table entry;
```

```
+-----+
| Table | Create Table
+-----+
| entry | CREATE TABLE `entry` (
  `memberID` int DEFAULT NULL,
  `TourID` int DEFAULT NULL,
  `year` year DEFAULT NULL,
  KEY `memberID` (`memberID`),
  KEY `TourID` (`TourID`),
  CONSTRAINT `entry_ibfk_1` FOREIGN KEY (`memberID`) REFERENCES `member` (`memberID`),
  CONSTRAINT `entry_ibfk_2` FOREIGN KEY (`TourID`) REFERENCES `tournament` (`TourID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
1 row in set (0.00 sec)
```

```
mysql> █
```

Task 4.6:

```
mysql> insert into type values('Associate', 60), ('Junior',150), ('Senior', 300), ('Social', 50);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
```

```
mysql> select * from type;
+-----+-----+
| type      | fee  |
+-----+-----+
| Associate | 60   |
| Junior    | 150  |
| Senior    | 300  |
| Social    | 50   |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> insert into tournament values(24, 'Leeston', 'Social'), (25, 'Kaiapoi', 'Social'), (36, 'WestCoast', 'Social'), (38, 'Canterburry', 'Open'), (40, 'Otago', 'Open');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
```

```
mysql> select * from tournament;
+-----+-----+-----+
| TourID | Tourname   | TourType |
+-----+-----+-----+
| 24     | Leeston    | Social   |
| 25     | Kaiapoi    | Social   |
| 36     | WestCoast  | Social   |
| 38     | Canterbury | Open     |
| 40     | Otago      | Open     |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> insert into team values('Team A', 'Tuesday', 239), ('Team B', 'Monday', 153);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> select * from team;
+-----+-----+-----+
| Teamname | practicenight | manager |
+-----+-----+-----+
| Team A   | Tuesday       | 239     |
| Team B   | Monday        | 153     |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> insert into member VALUES
-> (118, 'Mellisa', 'Mckenzie', 'Junior', '963270', 30, '2009-05-10', 153, 'F', NULL ),
-> (138, 'Michael', 'Stone', 'Senior', '983223', 30, '2013-05-13', 0, 'M', NULL ),
-> (153, 'Brenda', 'Nolen', 'Senior', '442749', 11, '2010-07-24', NULL, 'F', 'Team B' ),
-> (176, 'Helen', 'Branch', 'Social', '589419', 0, '2015-11-18', 0, 'F', NULL ),
-> (178, 'Sarah', 'Beck', 'Social', '226596', 0, '2014-01-06', 0, 'F', NULL ),
-> (228, 'Sandra', 'Burton', 'Junior', '244493', 26, '2015-06-21', 153, 'F', NULL ),
-> (235, 'William', 'Cooper', 'Senior', '7229454', 14, '2012-02-12', 153, 'M', 'Team B' ) ,
-> (239, 'Thomas', 'Spance', 'Senior', '797720', 10, '2010-06-04', 8, 'M', NULL ),
-> (258, 'Barbara', 'Olson', 'Senior', '370168', 16, '2015-07-11', 0, 'F', NULL ),
-> (286, 'Robert', 'Pollard', 'Junior', '617681', 19, '2015-07-26', 235, 'M', 'Team B' ),
-> (290, 'Thomas', 'Buxton', 'Senior', '268936', 26, '2012-07-10', 235, 'm', NULL ),
-> (323, 'Daniel', 'Wilcox', 'Senior', '665993', 3, '2013-04-30', 0, 'M', 'Team A' ),
-> (331, 'Thoma', 'Schmidt', 'Senior', '867492', 25, '2013-03-20', 153, 'M', NULL ),
-> (332, 'Deborah', 'Bridges', 'Senior', '279087', 12, '2011-03-05', 235, 'F', NULL ),
-> (339, 'Betty', 'Young', 'Senior', '597813', 21, '2013-03-30', 0, 'F', 'Team B' ),
-> (414, 'Jane', 'Gilmore', 'Junior', '459558', 5, '2011-05-12', 153, 'F', 'Team A' ),
-> (415, 'William', 'taylor', 'Senior', '137353', 7, '2011-11-09', 235, 'm', 'Team A' ),
-> (461, 'Robert', 'Reed', 'Senior', '994664', 3, '2009-07-18', 235, 'M', 'Team A' ),
-> (469, 'Carolyn', 'Willis', 'Junior', '688378', 29, '2014-12-27', 0, 'F', NULL ),
-> (487, 'Susan', 'Kent', 'Social', '707217', 0, '2014-09-29', 0, 'F', NULL )
-> ;
Query OK, 20 rows affected (0.01 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

```
mysql> select * from member;
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
118	Mellisa	Mckenzie	Junior	963270	30	2009-05-10	153	F	NULL
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
153	Brenda	Nolen	Senior	442749	11	2010-07-24	NULL	F	Team B
176	Helen	Branch	Social	589419	0	2015-11-18	0	F	NULL
178	Sarah	Beck	Social	226596	0	2014-01-06	0	F	NULL
228	Sandra	Burton	Junior	244493	26	2015-06-21	153	F	NULL
235	William	Cooper	Senior	7229454	14	2012-02-12	153	M	Team B
239	Thomas	Spance	Senior	797720	10	2010-06-04	8	M	NULL
258	Barbara	Olson	Senior	370168	16	2015-07-11	0	F	NULL
286	Robert	Pollard	Junior	617681	19	2015-07-26	235	M	Team B
290	Thomas	Buxton	Senior	268936	26	2012-07-10	235	m	NULL
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
332	Deborah	Bridges	Senior	279087	12	2011-03-05	235	F	NULL
339	Betty	Young	Senior	597813	21	2013-03-30	0	F	Team B
414	Jane	Gilmore	Junior	459558	5	2011-05-12	153	F	Team A
415	William	taylor	Senior	137353	7	2011-11-09	235	m	Team A
461	Robert	Reed	Senior	994664	3	2009-07-18	235	M	Team A
469	Carolyn	Willis	Junior	688378	29	2014-12-27	0	F	NULL
487	Susan	Kent	Social	707217	0	2014-09-29	0	F	NULL

```
20 rows in set (0.00 sec)
```

```
mysql> insert into entry VALUES
-> (118, 24 ,2013),
-> (228, 24 ,2014),
-> (228, 25 ,2014),
-> (228, 36 ,2014),
-> (235, 38 ,2012),
-> (235, 38 ,2014),
-> (235, 40 ,2013),
-> (235, 40 ,2014),
-> (239, 25 ,2014),
-> (239, 40 ,2012),
-> (258, 24 ,2013),
-> (258, 38 ,2013),
-> (286, 24 ,2012),
-> (286, 24 ,2013),
-> (286, 24 ,2014),
-> (415, 25 ,2014),
-> (415, 36 ,2013),
-> (415, 36 ,2014),
-> (415, 38 ,2012),
-> (415, 38 ,2014),
-> (415, 40 ,2012)
-> ;
Query OK, 21 rows affected (0.01 se
Records: 21 Duplicates: 0 Warning
```

```
mysql> select * from entry;
+-----+-----+-----+
| memberID | TourID | year |
+-----+-----+-----+
| 118 | 24 | 2013 |
| 228 | 24 | 2014 |
| 228 | 25 | 2014 |
| 228 | 36 | 2014 |
| 235 | 38 | 2012 |
| 235 | 38 | 2014 |
| 235 | 40 | 2013 |
| 235 | 40 | 2014 |
| 239 | 25 | 2014 |
| 239 | 40 | 2012 |
| 258 | 24 | 2013 |
| 258 | 38 | 2013 |
| 286 | 24 | 2012 |
| 286 | 24 | 2013 |
| 286 | 24 | 2014 |
| 415 | 25 | 2014 |
| 415 | 36 | 2013 |
| 415 | 36 | 2014 |
| 415 | 38 | 2012 |
| 415 | 38 | 2014 |
| 415 | 40 | 2012 |
+-----+-----+-----+
21 rows in set (0.00 sec)
```

Task 4.7:

Write the query for the following:

a) List the first name, last name, and phone numbers of all the members.

```
mysql> select member.firstName, member.lastName , member.Phone from member;
```

firstName	lastName	Phone
Mellisa	Mckenzie	963270
Michael	Stone	983223
Brenda	Nolen	442749
Helen	Branch	589419
Sarah	Beck	226596
Sandra	Burton	244493
William	Cooper	7229454
Thomas	Spance	797720
Barbara	Olson	370168
Robert	Pollard	617681
Thomas	Buxton	268936
Daniel	Wilcox	665993
Thoma	Schmidt	867492
Deborah	Bridges	279087
Betty	Young	597813
Jane	Gilmore	459558
William	taylor	137353
Robert	Reed	994664
Carolyn	Willis	688378
Susan	Kent	707217

```
20 rows in set (0.00 sec)
```

b) List complete information of all the male members.

```
mysql> select * from member where gender = 'M';
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
235	William	Cooper	Senior	7229454	14	2012-02-12	153	M	Team B
239	Thomas	Spance	Senior	797720	10	2010-06-04	8	M	NULL
286	Robert	Pollard	Junior	617681	19	2015-07-26	235	M	Team B
290	Thomas	Buxton	Senior	268936	26	2012-07-10	235	m	NULL
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
415	William	taylor	Senior	137353	7	2011-11-09	235	m	Team A
461	Robert	Reed	Senior	994664	3	2009-07-18	235	M	Team A

```
9 rows in set (0.00 sec)
```

c) List complete information of all the members who joined after 01-01-2013.

```
mysql> select * from member where JoinDate > '2013-01-01';
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
176	Helen	Branch	Social	589419	0	2015-11-18	0	F	NULL
178	Sarah	Beck	Social	226596	0	2014-01-06	0	F	NULL
228	Sandra	Burton	Junior	244493	26	2015-06-21	153	F	NULL
258	Barbara	Olson	Senior	370168	16	2015-07-11	0	F	NULL
286	Robert	Pollard	Junior	617681	19	2015-07-26	235	M	Team B
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
339	Betty	Young	Senior	597813	21	2013-03-30	0	F	Team B
469	Carolyn	Willis	Junior	688378	29	2014-12-27	0	F	NULL
487	Susan	Kent	Social	707217	0	2014-09-29	0	F	NULL

```
11 rows in set (0.00 sec)

mysql>
```

d) List name of all the members who belonged to Team A.

```
mysql> select * from member where team = 'Team A';
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
414	Jane	Gilmore	Junior	459558	5	2011-05-12	153	F	Team A
415	William	taylor	Senior	137353	7	2011-11-09	235	m	Team A
461	Robert	Reed	Senior	994664	3	2009-07-18	235	M	Team A

```
4 rows in set (0.00 sec)

mysql>
```

e) List complete information of all the senior members.

```
mysql> select * from member where memberType = 'Senior';
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
153	Brenda	Nolen	Senior	442749	11	2010-07-24	NULL	F	Team B
235	William	Cooper	Senior	7229454	14	2012-02-12	153	M	Team B
239	Thomas	Spance	Senior	797720	10	2010-06-04	8	M	NULL
258	Barbara	Olson	Senior	370168	16	2015-07-11	0	F	NULL
290	Thomas	Buxton	Senior	268936	26	2012-07-10	235	m	NULL
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
332	Deborah	Bridges	Senior	279087	12	2011-03-05	235	F	NULL
339	Betty	Young	Senior	597813	21	2013-03-30	0	F	Team B
415	William	taylor	Senior	137353	7	2011-11-09	235	m	Team A
461	Robert	Reed	Senior	994664	3	2009-07-18	235	M	Team A

```
12 rows in set (0.00 sec)

mysql>
```

f) List complete information of all the members in order of LastName.

```
mysql> select * from member order by LastName;
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
178	Sarah	Beck	Social	226596	0	2014-01-06	0	F	NULL
176	Helen	Branch	Social	589419	0	2015-11-18	0	F	NULL
332	Deborah	Bridges	Senior	279087	12	2011-03-05	235	F	NULL
228	Sandra	Burton	Junior	244493	26	2015-06-21	153	F	NULL
290	Thomas	Buxton	Senior	268936	26	2012-07-10	235	m	NULL
235	William	Cooper	Senior	7229454	14	2012-02-12	153	M	Team B
414	Jane	Gilmore	Junior	459558	5	2011-05-12	153	F	Team A
487	Susan	Kent	Social	707217	0	2014-09-29	0	F	NULL
118	Mellisa	Mckenzie	Junior	963270	30	2009-05-10	153	F	NULL
153	Brenda	Nolen	Senior	442749	11	2010-07-24	NULL	F	Team B
258	Barbara	Olson	Senior	370168	16	2015-07-11	0	F	NULL
286	Robert	Pollard	Junior	617681	19	2015-07-26	235	M	Team B
461	Robert	Reed	Senior	994664	3	2009-07-18	235	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
239	Thomas	Spance	Senior	797720	10	2010-06-04	8	M	NULL
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
415	William	taylor	Senior	137353	7	2011-11-09	235	m	Team A
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
469	Carolyn	Willis	Junior	688378	29	2014-12-27	0	F	NULL
339	Betty	Young	Senior	597813	21	2013-03-30	0	F	Team B

```
20 rows in set (0.01 sec)
```

g) Retrieve the number of records in Member table.

```
mysql> select count(*) from member;
```

count(*)
20

```
1 row in set (0.02 sec)
```

```
mysql>
```

h) Provide the first name and last name of the two coaches.

```
mysql> SELECT member.FirstName, member.LastName FROM member where memberId = 153 or memberId = 235;
```

FirstName	LastName
Brenda	Nolen
William	Cooper

```
2 rows in set (0.00 sec)
```

```
mysql>
```

i) Find the amount of fee provided by each member by mentioning member first name, last name, and fee. (Hint: use the member and type tables.)

```
mysql> select type.fee, member.FirstName, member.LastName from type, member where member.memberType = type.type;
```

fee	FirstName	LastName
150	Mellisa	Mckenzie
150	Sandra	Burton
150	Robert	Pollard
150	Jane	Gilmore
150	Carolyn	Willis
300	Michael	Stone
300	Brenda	Nolen
300	William	Cooper
300	Thomas	Spance
300	Barbara	Olson
300	Thomas	Buxton
300	Daniel	Wilcox
300	Thoma	Schmidt
300	Deborah	Bridges
300	Betty	Young
300	William	taylor
300	Robert	Reed
50	Helen	Branch
50	Sarah	Beck
50	Susan	Kent

```
20 rows in set (0.00 sec)

mysql>
```

j) Delete the record from Entry table where Member=415 and TourID=40.

```
mysql> DELETE FROM entry where memberID=415 and TourID=40.;
Query OK, 1 row affected (0.01 sec)
```

k) Update the Fee of Associate in Type table from 60 to 80.

```
mysql> update type set fee = 80 where type = 'Associate';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Task 4.8:

Numeric Functions:

CEIL (Ceiling): This function gives the smallest integer greater than or equal to a given numeric expression.

```
mysql> SELECT CEIL(5.3);
```

CEIL(5.3)
6

```
1 row in set (0.00 sec)

mysql>
```

COS (Cosine): Returns the cosine of an angle (in radians).

```
mysql> SELECT COS(0);
+-----+
| COS(0) |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)
```

DEGREES: Converts radians to degrees.

```
mysql> SELECT DEGREES(PI());
+-----+
| DEGREES(PI()) |
+-----+
|          180 |
+-----+
1 row in set (0.00 sec)
```

LOG10: Returns the base-10 logarithm of the argument.

```
mysql> SELECT LOG10(1000);
+-----+
| LOG10(1000) |
+-----+
|           3 |
+-----+
1 row in set (0.00 sec)

mysql>
```

MOD: Returns the remainder after dividing the first argument by the second argument.

```
mysql> SELECT MOD(10, 3);
+-----+
| MOD(10, 3) |
+-----+
|          1 |
+-----+
1 row in set (0.00 sec)

mysql> 
```

RADIANS: Converts degrees to radians.

```
mysql> SELECT RADIANS(180);
+-----+
| RADIANS(180) |
+-----+
| 3.141592653589793 |
+-----+
1 row in set (0.00 sec)
```

ROUND: Rounds a number to the nearest integer.

```
mysql> SELECT ROUND(5.7);
+-----+
| ROUND(5.7) |
+-----+
|          6 |
+-----+
1 row in set (0.00 sec)

mysql>
```

SQRT (Square Root): Returns the square root of the argument.

```
mysql> SELECT SQRT(25);
+-----+
| SQRT(25) |
+-----+
|          5 |
+-----+
1 row in set (0.00 sec)

mysql>
```

TRUNCATE: Truncates a number to a specified number of decimal places.

```
mysql> SELECT TRUNCATE(7.456, 2);
+-----+
| TRUNCATE(7.456, 2) |
+-----+
|          7.45 |
+-----+
1 row in set (0.00 sec)

mysql>
```

String Functions:

CONCAT: Concatenates two or more strings.

```
mysql> SELECT CONCAT('Ali', ' Asghar');
+-----+
| CONCAT('Ali', ' Asghar') |
+-----+
| Ali Asghar |
+-----+
1 row in set (0.00 sec)
```

UPPER: Converts a string to uppercase.

```
mysql> SELECT UPPER('Ali Asghar');
+-----+
| UPPER('Ali Asghar') |
+-----+
| ALI ASGHAR |
+-----+
1 row in set (0.00 sec)

mysql>
```

LOWER: Converts a string to lowercase.

```
mysql> SELECT LOWER('Ali Asghar');
+-----+
| LOWER('Ali Asghar') |
+-----+
| ali asghar          |
+-----+
1 row in set (0.00 sec)
```

REPEAT: Repeats a string a specified number of times.

```
mysql> SELECT REPEAT('abc', 3);
+-----+
| REPEAT('abc', 3) |
+-----+
| abcabcabc        |
+-----+
1 row in set (0.00 sec)
```

REVERSE: Reverses the characters in a string.

```
mysql> SELECT REVERSE('Ali Asghar');
+-----+
| REVERSE('Ali Asghar') |
+-----+
| rahgsA iLA            |
+-----+
1 row in set (0.00 sec)
```

```
mysql>
```

REGEXP: Checks if a string matches a regular expression.

```
mysql> SELECT 'apple' REGEXP '^a';
+-----+
| 'apple' REGEXP '^a' |
+-----+
| 1 |
+-----+
1 row in set (0.03 sec)

mysql>
```

REPLACE: Replaces occurrences of a specified string.

```
mysql> SELECT REPLACE('Ali, Ahmad', 'Ahmad', 'Asghar');  
+-----+  
| REPLACE('Ali, Ahmad', 'Ahmad', 'Asghar') |  
+-----+  
| Ali, Asghar                               |  
+-----+  
1 row in set (0.00 sec)
```

LENGTH: Returns the length of a string in bytes.

```
mysql> SELECT LENGTH('Hello');
+-----+
| LENGTH('Hello') |
+-----+
|                5 |
+-----+
1 row in set (0.00 sec)
```

LTRIM (Left Trim): Removes leading spaces from a string.

```
mysql> SELECT LTRIM('   Trimmed');
+-----+
| LTRIM('   Trimmed') |
+-----+
| Trimmed              |
+-----+
1 row in set (0.00 sec)
```

RTRIM (Right Trim): Removes trailing spaces from a string.

```
mysql> SELECT RTRIM('Trimmed   ');
+-----+
| RTRIM('Trimmed   ') |
+-----+
| Trimmed              |
+-----+
1 row in set (0.00 sec)
```

Date & Time Functions:

CURDATE: Returns the current date.

```
mysql> SELECT CURDATE();
+-----+
| CURDATE() |
+-----+
| 2024-04-21 |
+-----+
1 row in set (0.00 sec)
```

WEEK: Returns the week number for a date.

```
mysql> SELECT WEEK('2024-04-21');
+-----+
| WEEK('2024-04-21') |
+-----+
| 16 |
+-----+
1 row in set (0.00 sec)
```

DATE_FORMAT: Formats a date as specified.

```
mysql> SELECT DATE_FORMAT(NOW(), '%Y-%m-%d %H:%i:%s');
+-----+
| DATE_FORMAT(NOW(), '%Y-%m-%d %H:%i:%s') |
+-----+
| 2024-04-21 17:20:50 |
+-----+
1 row in set (0.00 sec)
```

QUARTER: Returns the quarter from a date.

```
mysql> SELECT QUARTER('2024-04-21');
+-----+
| QUARTER('2024-04-21') |
+-----+
| 2 |
+-----+
1 row in set (0.00 sec)
```

NOW: Returns the current date and time.

```
mysql> SELECT NOW();
+-----+
| NOW() |
+-----+
| 2024-04-21 17:22:25 |
+-----+
1 row in set (0.00 sec)

mysql>
```

SYSDATE: Returns the time at which the function executes.

```
mysql> SELECT SYSDATE();
+-----+
| SYSDATE() |
+-----+
| 2024-04-21 17:22:53 |
+-----+
1 row in set (0.00 sec)
```

Task 4.9:

Comparison:

!=

```
mysql> select * from member where coach != 0;
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
118	Mellisa	Mckenzie	Junior	963270	30	2009-05-10	153	F	NULL
228	Sandra	Burton	Junior	244493	26	2015-06-21	153	F	NULL
235	Wiliam	Cooper	Senior	7229454	14	2012-02-12	153	M	Team B
239	Thomas	Spance	Senior	797720	10	2010-06-04	8	M	NULL
286	Robert	Pollard	Junior	617681	19	2015-07-26	235	M	Team B
290	Thomas	Buxton	Senior	268936	26	2012-07-10	235	m	NULL
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
332	Deborah	Bridges	Senior	279087	12	2011-03-05	235	F	NULL
414	Jane	Gilmore	Junior	459558	5	2011-05-12	153	F	Team A
415	William	taylor	Senior	137353	7	2011-11-09	235	m	Team A
461	Robert	Reed	Senior	994664	3	2009-07-18	235	M	Team A

11 rows in set (0.00 sec)

>

```
mysql> select * from member where JoinDate > '2012-01-01'
-> ;
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
176	Helen	Branch	Social	589419	0	2015-11-18	0	F	NULL
178	Sarah	Beck	Social	226596	0	2014-01-06	0	F	NULL
228	Sandra	Burton	Junior	244493	26	2015-06-21	153	F	NULL
235	William	Cooper	Senior	7229454	14	2012-02-12	153	M	Team B
258	Barbara	Olson	Senior	370168	16	2015-07-11	0	F	NULL
286	Robert	Pollard	Junior	617681	19	2015-07-26	235	M	Team B
290	Thomas	Buxton	Senior	268936	26	2012-07-10	235	m	NULL
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
339	Betty	Young	Senior	597813	21	2013-03-30	0	F	Team B
469	Carolyn	Willis	Junior	688378	29	2014-12-27	0	F	NULL
487	Susan	Kent	Social	707217	0	2014-09-29	0	F	NULL

13 rows in set (0.00 sec)

Boolean:

AND

```
mysql> select * from member where gender = 'F' and memberType = 'Junior';
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
118	Mellisa	Mckenzie	Junior	963270	30	2009-05-10	153	F	NULL
228	Sandra	Burton	Junior	244493	26	2015-06-21	153	F	NULL
414	Jane	Gilmore	Junior	459558	5	2011-05-12	153	F	Team A
469	Carolyn	Willis	Junior	688378	29	2014-12-27	0	F	NULL

```
4 rows in set (0.00 sec)

mysql>
```

OR

```
mysql> select * from member where gender = 'M' or memberType = 'Junior';
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
118	Mellisa	Mckenzie	Junior	963270	30	2009-05-10	153	F	NULL
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
228	Sandra	Burton	Junior	244493	26	2015-06-21	153	F	NULL
235	William	Cooper	Senior	7229454	14	2012-02-12	153	M	Team B
239	Thomas	Spance	Senior	797720	10	2010-06-04	8	M	NULL
286	Robert	Pollard	Junior	617681	19	2015-07-26	235	M	Team B
290	Thomas	Buxton	Senior	268936	26	2012-07-10	235	m	NULL
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
414	Jane	Gilmore	Junior	459558	5	2011-05-12	153	F	Team A
415	William	taylor	Senior	137353	7	2011-11-09	235	m	Team A
461	Robert	Reed	Senior	994664	3	2009-07-18	235	M	Team A
469	Carolyn	Willis	Junior	688378	29	2014-12-27	0	F	NULL

```
13 rows in set (0.00 sec)

mysql>
```

Special Operators:

distinct:

```
mysql> select distinct team from member;
```

team
NULL
Team A
Team B

```
3 rows in set (0.19 sec)
```

between:

```
mysql> select * from member where JoinDate between '2013-01-01' and '2014-01-01';
```

memberID	FirstName	LastName	memberType	Phone	Handicap	JoinDate	coach	gender	team
138	Michael	Stone	Senior	983223	30	2013-05-13	0	M	NULL
323	Daniel	Wilcox	Senior	665993	3	2013-04-30	0	M	Team A
331	Thoma	Schmidt	Senior	867492	25	2013-03-20	153	M	NULL
339	Betty	Young	Senior	597813	21	2013-03-30	0	F	Team B

```
4 rows in set (0.01 sec)
```

Task 4.10:

a) Add new column DOB to store member date of birth. Its type is date and can be null.

```
mysql> ALTER TABLE member ADD dob DATE;
Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> describe member;
```

Field	Type	Null	Key	Default	Extra
memberID	int	NO	PRI	NULL	
FirstName	varchar(20)	YES		NULL	
LastName	varchar(20)	YES		NULL	
memberType	varchar(20)	YES	MUL	NULL	
Phone	int	YES	UNI	NULL	
Handicap	int	YES		NULL	
JoinDate	date	YES		NULL	
coach	int	YES	MUL	NULL	
gender	varchar(10)	YES		NULL	
team	varchar(20)	YES	MUL	NULL	
dob	date	YES		NULL	

```
11 rows in set (0.01 sec)
```

b) Now change the name of newly added column from DOB to M_DOB with date as data type and not null.

```
mysql> ALTER TABLE member CHANGE dob DOB_M DATE not null;
Query OK, 20 rows affected, 20 warnings (0.14 sec)
Records: 20 Duplicates: 0 Warnings: 20
```

```
mysql> describe member;
```

Field	Type	Null	Key	Default	Extra
memberID	int	NO	PRI	NULL	
FirstName	varchar(20)	YES		NULL	
LastName	varchar(20)	YES		NULL	
memberType	varchar(20)	YES	MUL	NULL	
Phone	int	YES	UNI	NULL	
Handicap	int	YES		NULL	
JoinDate	date	YES		NULL	
coach	int	YES	MUL	NULL	
gender	varchar(10)	YES		NULL	
team	varchar(20)	YES	MUL	NULL	
DOB_M	date	NO		NULL	

```
11 rows in set (0.01 sec)
```

c) Now drop the M_DOB column from member table.

```
mysql> ALTER TABLE member DROP COLUMN DOB_M;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> describe member;
```

Field	Type	Null	Key	Default	Extra
memberID	int	NO	PRI	NULL	
FirstName	varchar(20)	YES		NULL	
LastName	varchar(20)	YES		NULL	
memberType	varchar(20)	YES	MUL	NULL	
Phone	int	YES	UNI	NULL	
Handicap	int	YES		NULL	
JoinDate	date	YES		NULL	
coach	int	YES	MUL	NULL	
gender	varchar(10)	YES		NULL	
team	varchar(20)	YES	MUL	NULL	

```
10 rows in set (0.01 sec)
```

```
mysql>
```

d) Next drop the primary key TourID from tournament table.

```
mysql> ALTER TABLE tournament DROP COLUMN TourID;
Query OK, 5 rows affected (0.06 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
```

e) Now add new primary key TourID into tournament table.

```
mysql> ALTER TABLE entry ADD PRIMARY KEY (TourID);
ERROR 1062 (23000): Duplicate entry '24' for key 'entry.PRIMARY'
mysql> show create table entry;
+-----+-----+
| Table | Create Table
+-----+-----+
| entry | CREATE TABLE `entry` (
  `memberID` int DEFAULT NULL,
  `TourID` int DEFAULT NULL,
  `year` year DEFAULT NULL,
  KEY `memberID` (`memberID`),
  KEY `TourID` (`TourID`),
  CONSTRAINT `entry_ibfk_1` FOREIGN KEY (`memberID`) REFERENCES `member` (`memberID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
```

f) Next drop the foreign key Coach from member table.

```
mysql> ALTER TABLE member DROP FOREIGN KEY member_ibfk_3;  
Query OK, 0 rows affected (0.04 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> _
```

g) Now add the new foreign key Coach from member table.

```
mysql> ALTER TABLE member ADD FOREIGN KEY(Coach) REFERENCES member(memberID);  
Query OK, 0 rows affected (0.03 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> _
```

Registration #: _____

Name: _____

Date: _____

CSE 404L – Database Management Systems Lab**LAB ASSESSMENT RUBRICS****DBMS LAB 04 – Introduction to MySQL**

Dimension	Exemplary	Acceptable	Developing	Unsatisfactory	Student Score out of 10 Marks
	10	8	6	4	
Overall Impression of Lab Report	Report is complete, well written, and organized appropriately with additional elements that enhance it.	Report is complete, briefly written, and organized. Lacks additional elements.	Report is mostly complete, loosely written, and fairly organized.	Report is incomplete, sloppy, and/or disorganized.	
Submission	Report is submitted on time.	Report is submitted within 24 hours of due date.	Report is submitted within 72 hours of due date.	Report was more than 3 days overdue.	
Specification	Tasks work and exceed specifications.	Tasks work and meet all specifications.	Tasks work and meet partial specifications.	Tasks work but fail to meet any specification.	
Output Figures/Graphics	All the output figures and graphics are shown clearly and labeled.	Most output figures and graphics are shown clearly and labeled.	Few of the output figures and graphics are shown and labeled.	Output figures and graphics are not shown and not labeled.	
Verbal Communication and Understanding	Answered clearly and accurately with sufficient knowledge of MySQL.	Answered clearly and accurately with average knowledge of MySQL.	Answered with limited knowledge of MySQL.	Tried to answer but failed to show any knowledge of MySQL.	

Marks: (_____ + _____ + _____ + _____ + _____)/5 = _____

Teacher Remarks and Signature: _____